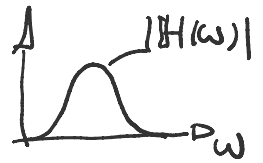


8.59) NÄGÅT TIPS

$$|H(\omega)| = \frac{\omega R_C}{1 + (\omega R_C)^2}$$

MAX INTRÄFFAR DÄR $\frac{d}{d\omega} |H(\omega)| = 0$

DERIVERA

$$\frac{d}{d\omega} |H(\omega)| = \frac{R_C (1 + (\omega R_C)^2) - \omega R_C (2\omega R_C) R_C}{(1 + (\omega R_C)^2)^2}$$

$$\Rightarrow R_C (1 + (\omega R_C)^2) = \omega R_C (2\omega R_C) R_C$$

$$1 + (\omega R_C)^2 - 2\omega^2 R_C^2 = 0$$

$$\omega^2 = \frac{1}{(R_C)^2}$$

$$\omega = \frac{1}{R_C} = \omega_0$$

Bredbandsnivå då $|H(\omega)|_{\max} \frac{1}{\sqrt{2}} = |H(\omega)|$

$$|H(\omega)|_{\max} = \frac{\omega_0 R_C}{1 + (\omega_0 R_C)^2} = \frac{1}{1 + 1} = \frac{1}{2}$$

$$\Rightarrow \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\omega R_C}{1 + (\omega R_C)^2} \quad \text{Lös för } \omega$$

$$1 + (\omega R_C)^2 = 2\sqrt{2} \omega R_C$$

$$\omega^2 - \frac{2\sqrt{2}}{R_C} \omega + \frac{1}{(R_C)^2} = 0$$

$$\begin{aligned} \omega_{1,2} &= \frac{\sqrt{2}}{R_C} \pm \sqrt{\frac{2}{(R_C)^2} - \frac{1}{(R_C)^2}} = \\ &= \frac{\sqrt{2}}{R_C} \pm \frac{1}{R_C} = \begin{cases} 2.41 \frac{1}{R_C} \\ 0.41 \frac{1}{R_C} \end{cases} \end{aligned}$$

$$= \frac{\sqrt{c}}{R_c} \pm \frac{1}{R_c} = \left\{ 0.41 \frac{1}{R_c} \right. \quad \sim$$

